Sparky Redeployment Guide

Nov 6, 2026 · @Orange

Start here

Use **Route A** unless something stops you: on this project's connection, it took about 15 minutes from deleted to answering.

Your situation	Use	Time, once Docker is installed	Needs
New laptop or lab computer with good internet	**A: Docker from GitHub**	About 15 min on the test connection, longer on slower ones	Docker Desktop, GitHub token, about 25 GB free
Slow or blocked internet	**B: USB bundle**	Several minutes to load, not yet timed	Docker Desktop, the USB bundle, about 25 GB free
Docker can't be installed, or no admin rights for it	**C: Manual install**	About an hour or more	Admin rights may still be needed, plus your Drive files
You can't install anything	**Bring your own laptop**	None	The laptop already set up

Have these ready before you start:

- Your GitHub login, for the `ORANGElemoen` account
- Your ngrok login, only if you want the internet route
- Your Google Drive login, for backups and voice training files
- Sparky and its USB cable
- For a lab: permission to install software, or Docker already installed

Proven and not yet proven (6 Nov 2026):

Tested	Result
Delete everything from Docker and restore from GitHub, on the same laptop	Worked, with the app tested from the laptop

Tested	Result
Docker with no internet, with Sparky answering	Worked
Sparky connecting to Docker with its saved key	Worked
A genuinely new computer	**Not yet tested**
Computer with no graphics card	**Not yet tested**, so speed is unknown
USB bundle (Route B)	**Not yet tested**
A real lab computer	**Not yet tested**, so try one before relying on it

What is stored where

Sparky survives losing this laptop only if these copies stay up to date, so check them against this table.

What	Where	Link or location	If lost
Code, firmware, circuit and enclosure files, Docker recipe, this project's docs	GitHub repo, public	ORANGElemoen/SPARKY_RAGdashboard	Your laptop's copy can restore it
The full app, ready to run	GitHub images, private	`ghcr.io/orangelemoen/sparky` and `ghcr.io/orangelemoen/sparky-ollama`, listed on your packages page	Rebuild from the repo and a backup zip
Sheets, voice lines, faces, Sparky's voice, database, question log	Backup zip on Google Drive	`sparky_backup_<date>_<time>.zip`, made by `scripts/backup_sparky.py`	The image holds a copy from its build date
Voice training files	Google Drive	`MyDrive/sparky_voice/`, with `latest.ckpt`, `reference.wav` and more	Only needed to retrain the voice

What	Where	Link or location	If lost
Firmware's private settings: `local_config.h`	Backup zip	`firmware/sparky/local_config.h`	Recreate it: see Route C

Never stored in the repo, images or backups:

- Your ngrok token and dashboard login: set up again on each computer
- Your GitHub token: make a new one when needed
- Wi-Fi passwords: saved only on the board

The images are a snapshot. They hold your content as it was on the last build, currently 6 Nov 2026. Changes made after that are only in the backup zip until you update the images.

Keep the images private. They hold the question log and Sparky's voice. A real deployment with children must leave the question log out, as your report should explain.

Route A: Docker from GitHub

This gets you the whole app with your content, using one click after the first download.

Needs: Windows 10 or 11 with virtualisation on, 16 GB of memory ideally, about 25 GB free, admin rights for the install.

1. **Install Docker Desktop** from docker.com, and restart Windows if asked. It sets up WSL2 for you. If Windows says virtualisation is off, it needs enabling in the BIOS.
2. **Open Docker Desktop once,** and accept the licence. You can skip signing in. Wait until it says it's running.
3. **Get the project files.** On the repo page, click **Code**, then **Download ZIP**, and unzip it. With Git installed, this works too:

   ```
   git clone https://github.com/ORANGElemoen/SPARKY_RAGdashboard.git
   ```

4. **Make a token** at github.com/settings/tokens/new, signed in as `ORANGElemoen`. Tick only `read:packages`, and set an expiry of 7 days for a lab. Leave the page open.
5. **Log Docker in.** Open PowerShell in the project folder, then run this line. When it says `Press Enter once it is copied`, copy the token from GitHub, return to PowerShell and press Enter.

```
powershell -ExecutionPolicy Bypass -File scripts\docker_login.ps1
```

It should end with **Login Succeeded**. Don't paste the token into a password prompt yourself, as that silently failed in the VS Code terminal.

6. **Start Sparky.** Double-click `scripts\docker_start.bat`. The first time, it downloads about 13 GB, which took 13.6 minutes on the test connection.
7. **Answer the internet question.** Press Enter to run local only, or type **Y** and enter your ngrok token and a dashboard login. You can do this later with **N**.
8. **Wait for the control window** to show **OK** on each row, about a minute. **Board** turns **OK** once Sparky connects, and **Internet** only if you set it up.
9. Reconnect Sparky, then check it worked.

To start it next time: double-click `scripts\docker_start.bat`. Nothing downloads again.

The control window's keys:

Key	What it does
O	Opens the dashboard
C	Copies the internet address
T	Retries the tunnel
N	Sets up internet access
F	Forgets your ngrok token and login
S	Stops Sparky, keeping your data. Closing the window leaves it running

Route B: USB bundle

This needs no internet at the other end, but **it hasn't been tested yet**, and you must make the bundle before you need it. Docker Desktop still has to be installed, so download its installer onto the same stick.

Make the bundle in advance on a computer that already has the images, using a drive with at least 18 GB free. A 32 GB stick is plenty. Replace `E:\` with your drive:

```
powershell -ExecutionPolicy Bypass -File scripts\docker_bundle.ps1 E:\
```

This creates a `Sparky-Docker` folder holding the images, the compose files and the scripts. No ngrok token or login goes in, but your content does, so keep the stick private.

On the new computer:

1. Install Docker Desktop from the installer on the stick, and restart Windows if asked.
2. Open Docker Desktop once, accept the licence and wait until it's running.
3. Copy the `Sparky-Docker` folder to the computer's own drive, which should be faster than running from USB.
4. Double-click `scripts\docker_start.bat` in that folder. It loads `sparky-images.tar` once, then starts Sparky.
5. Press Enter at the internet question, as there's no internet to use.
6. Reconnect Sparky, then check it worked.

Remake the bundle whenever you change sheets, voice lines or faces, because it's a snapshot too.

Route C: Manual install without Docker

Use this when Docker isn't an option, or to set up a laptop for development and firmware work. It's how this laptop is set up.

1. **Install the tools** in PowerShell, approving any Windows prompts:

   ```
   winget install --id Git.Git --exact
   winget install --id Python.Python.3.14 --exact
   winget install --id Ollama.Ollama --exact
   winget install --id Gyan.FFmpeg --exact
   winget install --id ArduinoSA.CLI --exact
   winget install --id 9MVS1J51GMK6 --source msstore
   ```

   The last one is ngrok, and it's only needed for the internet route. Use the Store version, because Windows Defender blocked the direct download. Arduino CLI is only for firmware work.
2. **Open a new terminal,** so the new tools are found, then get the code and its packages:

```
cd C:\Users\<you>
git clone https://github.com/ORANGElemoen/SPARKY_RAGdashboard.git
cd SPARKY_RAGdashboard
python -m venv .venv
.\.venv\Scripts\python.exe -m pip install -r requirements.txt -r
deployment\requirements\voice_requirements.txt
```

3. **Pull the AI model:** `ollama pull qwen3.5:4b`, about 3.3 GB.
4. **Restore your content.** Download the newest `sparky_backup_...zip` from Google Drive, and unzip it into the `SPARKY_RAGdashboard` folder, keeping its folders, so `data\rag_database.db` lands in place. Do this while the server is stopped.
   - Original PDF downloads only work if the project is at the same path as before, `C:\Users\chris\SPARKY_RAGdashboard`. Answers work either way.
5. **Go online once for the first question,** so the speech and search models download. After that, it works with no internet.
6. **Start it:** double-click `scripts\start_sparky_server.bat`. It starts Ollama, the hotspot, the server and the control window.
7. **Set up the internet route, if you want it:** press **N** in the control window.
8. Reconnect Sparky, then check it worked.

If `local_config.h` is lost, recreate `firmware/sparky/local_config.h`, using your address from the ngrok dashboard. Never commit this file, as the repo is public:

```
#pragma once
#define SPARKY_SERVER "http://192.168.137.1:8001"
#define SPARKY_INTERNET "https://<your-address>.ngrok-free.dev"
```

If Sparky's voice is missing, the server falls back to a stock voice. Get `en_US-sparky-medium.onnx` from `MyDrive/sparky_voice/` and put it in `data\piper_voices\`.

Reconnect Sparky

Your existing board doesn't need flashing: it only needs to join a Wi-Fi network it can use, and a new computer's hotspot counts as a new network.

Set the new computer's hotspot first. In Windows, go to **Settings > Network & internet > Mobile hotspot > Edit**, and set the band to **2.4 GHz**, because the ESP32 can't use 5 GHz. Note the name and password. The start scripts turn the hotspot on, but they don't set this.

Then choose your route:

Setup at the venue	What Sparky joins	Route it uses	Voice quality
Computer with you, hotspot works	The computer's hotspot	Local, no internet needed	Full
Computer with you, hotspot blocked	Your phone's hotspot or the venue's Wi-Fi	ngrok	Reduced, like a phone call
Computer left elsewhere, running	Your phone's hotspot or the venue's Wi-Fi	ngrok	Reduced, like a phone call

To join a new network:

1. Press Sparky's button **5 times quickly**. It chirps, restarts and opens its setup page for 3 minutes. It also opens by itself at startup when no remembered network is around.
2. On a phone or tablet, join the `Sparky Setup` Wi-Fi, password `sparky123`. The page should open by itself. If not, go to `192.168.4.1`.
3. Choose the network and enter its password. Leave everything else alone, then save.

Sparky remembers up to 8 networks and prefers one where the local server answered. It can't join networks needing a username or a sign-in page, such as eduroam.

Check these only if something's wrong:

- **Local server address** should be `http://192.168.137.1:8001`.
- **Internet address** must match the ngrok account in use. A different account means a different address.
- **Device key:** your board's key works with Routes A and B, and with C after restoring a backup. A fresh database needs a new one: on the dashboard's **Devices** page, click **Register device**, then enter the key on the setup page.

Button summary:

Presses	Result
Hold, speak, let go	Asks a question
3 quick	Says goodbye and sleeps. Any press wakes it
5 quick	Opens the setup page

To flash firmware, which needs the manual tools from Route C and `local_config.h`. A new board, or a firmware change, is when this applies:

```
arduino-cli core install esp32:esp32
arduino-cli lib install WiFiManager ArduinoJson MD_MAX72XX
arduino-cli board list
arduino-cli compile --fqbn esp32:esp32:esp32:PartitionScheme=no_ota --clean
firmware/sparky
arduino-cli upload --fqbn esp32:esp32:esp32:PartitionScheme=no_ota -p COM3
firmware/sparky
```

Replace `COM3` with the port `board list` shows. If no port appears, the CH340 USB driver may be missing, though Windows installed it by itself here. Before uploading, press and hold **BOOT**, tap **EN**, and keep holding until it finishes. If it stops at `Wrong boot mode detected (0x13)`, just try again, as about every other attempt failed that way. Always use the `no_ota` option, because any other layout wipes the voice lines. They then re-download from the server.

Check it worked

A redeployment is done when Sparky answers a sheet question out loud, so run these before you rely on it. The first answer can be slow, so ask one warm-up question before a demo.

- [ ] The control window shows **OK** for AI model, Server and Hotspot
- [ ] **Board** shows **OK** after Sparky connects
- [ ] Pressing **O** opens the dashboard, and the document list shows your 13 sheets or more
- [ ] Sparky Studio, at `/admin/studio/view`, shows your voice lines and the SPARKY face
- [ ] Restarting Sparky plays "Sparky here!" and then "Wi-Fi connected!"
- [ ] "How do I make a lemon battery?" gets a spoken answer based on the sheet, with the wall-socket warning
- [ ] "Why is the sky blue?" still gets an answer, from general knowledge
- [ ] If you set up the internet route: **Internet** shows **OK**, and the address asks for your login on a phone using mobile data
- [ ] For a demo with no internet: unplug everything and repeat the lemon question

Expected timing on a laptop with a graphics card: the model loads in about a minute, and answers start in about 4 to 10 seconds over the hotspot. Over ngrok, expect about 8 to 14 seconds and a thinner voice.

Lab computers

Ask the lab these questions before the day, because any of them can rule out a route. Bringing your own laptop is the safe fallback if the answers are unclear.

Ask	Why it matters	If the answer is no
Can you install Docker Desktop, or is it already there?	Needs admin rights and virtualisation	Bring your laptop
Is there an NVIDIA graphics card?	Without one, the AI runs on the processor. Speed is untested and may be much slower	Rehearse first, or bring your laptop
Is there about 25 GB free, and at least 16 GB of memory?	The images and models need it	Bring your laptop
Is there a Wi-Fi card, and is the hotspot allowed?	Sparky uses it for the local route	Use a phone hotspot and ngrok
Are ngrok, GitHub and Docker's download site allowed through the network?	Route A and the internet route need them	Use Route B for the images, and the local route for Sparky

Before leaving a shared computer:

1. Press **F** in the control window to remove your ngrok token and login.
2. Press **S** to stop Sparky.
3. Run `docker logout ghcr.io` to remove your GitHub login.
4. Delete what you installed, as Sparky's images hold your content and the question log:

```
docker compose --profile internet down -v
docker image rm sparky:latest sparky-ollama:latest ngrok/ngrok:latest
ghcr.io/orangelemoen/sparky:latest ghcr.io/orangelemoen/sparky-
ollama:latest
```

5. Delete the project folder.
6. Delete the token at github.com/settings/tokens if it's still active, and reset your ngrok token if you're unsure who could have seen it.

If everything is lost

Losing this laptop costs you at most your changes since the last backup, because GitHub and Google Drive hold everything else. Follow Route A on any computer first, then come back here for anything missing.

What's gone	How to recover
The laptop	Route A gets Sparky running. Route C sets up a new development laptop
Your newest sheets, voice lines or faces	Unzip the newest backup into a Route C setup. Then rebuild and push the images
GitHub images	Rebuild them, using the repo and the newest backup: see below
Sparky's trained voice	It's in the backup zip and in the image. Otherwise `MyDrive/sparky_voice/`, and `latest.ckpt` can export it again without full retraining
GitHub account access	Google Drive still holds your data. Use a local copy of the code if you have one, and recover the account through GitHub
The board	Flash a new one, using Reconnect Sparky. Wiring is in `docs/sparky_wiring_guide.pdf`, and `hardware/` holds the circuit and enclosure files

To rebuild the images from nothing: first complete Route C steps 1, 2 and 4, then log in with a token that has `write:packages` ticked, and run:

```
scripts\docker_build.bat
docker tag sparky:latest ghcr.io/orangelemoen/sparky:latest
docker tag sparky-ollama:latest ghcr.io/orangelemoen/sparky-ollama:latest
docker push ghcr.io/orangelemoen/sparky:latest
docker push ghcr.io/orangelemoen/sparky-ollama:latest
```

The first build took more than 30 minutes and needs tens of GB free while it runs.

Troubleshooting

These are problems we actually hit, with what fixed them.

What you see	Cause	Fix
`docker login` says `denied` or `password is empty`	The hidden paste received nothing in the VS Code terminal	Use `scripts\docker_login.ps1`
`docker push` says `denied` near the end of a long upload	Probably the upload's login expired. The layers usually arrive anyway	Run the same push again
Hotspot says it can't start with no connection	The Settings page refuses with no internet	Use the start scripts, which work around it
Sparky doesn't appear on the hotspot, or can't find it	The hotspot is on 5 GHz	Set the band to 2.4 GHz
"Oops, I can't get an answer" just after a server restart	Sparky switched to its backup route	Wait about 25 seconds and ask again. It normally retries by itself
First answer takes 30 to 60 seconds	The AI model is loading, or warming up	Wait for the control window, and ask a warm-up question
Answers stutter over ngrok	The old full-quality audio needs 44 KB/s, against about 22 KB/s available	Current firmware compresses it. Reflash if yours doesn't

What you see	Cause	Fix
Upload stops at `Wrong boot mode detected (0x13)`	BOOT wasn't held when the board reset	Hold BOOT, tap EN, keep holding, try again
Docker's file has grown huge	It never shrinks by itself	See below
`Internet` shows **WAIT** and stays there	The tunnel can't start, or the token or network is the problem	Press **T**. Check with `docker compose --profile internet logs ngrok`
Windows Defender blocks ngrok	The direct download gets flagged	Install the Microsoft Store version, or use Docker's route

To shrink Docker's disk file, run `docker builder prune -a -f`, stop Docker Desktop, run `wsl --shutdown`, then compact the file with diskpart as an administrator:

```
select vdisk file="C:\Users\
<you>\AppData\Local\Docker\wsl\disk\docker_data.vhdx"
attach vdisk readonly
compact vdisk
detach vdisk
```

That took it from 44.5 GB to 20.2 GB here.

Keeping it recoverable

This guide only works if the copies stay current, so do this after any change you'd hate to lose.

When	Do this	Time
After changing sheets, voice lines or faces	Run `.\.venv\Scripts\python.exe scripts\backup_sparky.py`, then upload the zip from `backups\` to Google Drive	A few minutes
After changing code	Commit and push to GitHub	A few minutes
Before relying on Docker with new content	Run `scripts\docker_build.bat`, tag and push the images as in If everything is lost	30 minutes or more

When	Do this	Time
Before a lab visit	Try the check list, and remake the USB bundle if you use one	Under an hour
When your GitHub token expires	Make a new one when you next need it	A minute

A computer already running Sparky keeps its own content after you update the images. To replace it with the new build, run `docker compose down -v`, which **deletes any changes made on that computer**.

GitHub currently lists private image storage as free and promises at least a month's notice before charging. If that changes, your repo and backup zip can still rebuild everything.